Université Concordia University

INSTITUTE FOR CO-OPERATIVE EDUCATION

LOUAY DRIDI

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EDUCATION

Bachelor of Engineering – Software Engineering Co-op **2024 - 2028**
Concordia University, Montreal, QC

- GPA: 4.0 / 4.3, Dean's List student

WORK EXPERIENCE

Full Stack Software Engineer **Sep 2026 – now**
Clutch-VR, Victoria, British Columbia

- Engineered cross-platform data flows between **React/TypeScript, Unity/C#, and Firebase Realtime Database**, implementing real-time synchronization, role-based access, and server-side validation across web and VR clients.
- Built and deployed production features using **React, Node.js, Firebase, and Vercel**, implementing **REST endpoints, CI/CD workflows and environment-based configuration** across development and production environment.

Infrastructure Software Engineer **May 2026 – Aug 2026**
Genetec, Montreal, Quebec

- Led the implementation of an **EPM** by migrating Security Center's embedded browser infrastructure from **CefSharp** to **Microsoft Edge WebView2** in collaboration with cross-functional teams.
- Debugged **multi-threaded C#/.NET** services by analyzing **thread synchronization**, **async execution**, and **concurrency-related defects**.
- Performed **STRIDE-based threat modeling** and **security impact assessments**, identifying attack vectors, trust boundaries, and authentication vulnerabilities.

VR Software Developer **Aug 2025 – May 2026**
Clutch-VR, Victoria, British Columbia

- Lead VR application development using **Unity** and **C#** and improving design-review efficiency by **80%**.
- Architected **Firebase Realtime Database** and **Authentication** to enable data synchronization across sessions.
- Trained a **YOLOv11** model for football play detection using **PyTorch** and **OpenCV** for preprocessing, achieving a **96%** mAP and reducing manual setup time by **60%**.

PERSONAL PROJECTS

Neural Network from Scratch | *Source Code*

- Developed a multilayer neural network in **Python/NumPy** using matrix operations, forward propagation, **backpropagation, and gradient descent**.
- Implemented **ReLU/Softmax, cross-entropy loss, and mini-batch training**, with performance tracking and visualization using **Matplotlib**.

Sudoku Solver Engine App | *Source Code* | App (Windows only)

- Developed a logic-based Sudoku solver in **C** using constraint propagation across different sections.
- Integrated the solver into a **C#** and **.NET Windows Forms** app leveraging **sudoku.dll** via **P/invoke**.

Face Recognition | *Source Code*

- Facial recognition in **Python** using **OpenCV** and **machine learning KNN** algorithm.
- Save, label, and manage facial data for future recognition with features for tracking recognition.

TECHNICAL KNOWLEDGE

- **Libraries/Frameworks** | ReactJS, GIT, NodeJS, .NET, PostgreSQL, PyTorch, OpenCV, Flutter
- **Programming** | C/C++ • C# • Java • Python • HTML • CSS • JS • PHP • Unity • Unreal Engine